

Recursion

Example problem

↳ fibonacci

0 1 1 2 3 5 8 13 - - - - -

$$F(0)=1 \quad F(1)=1 \quad n>1 \Rightarrow F(n)=F(n-1)+F(n-2)$$

def $F(n)$:

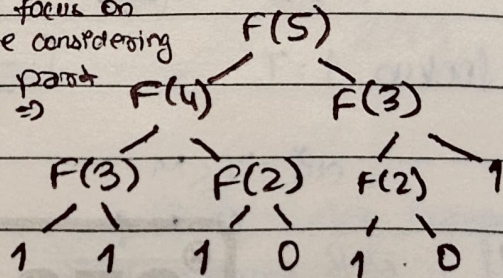
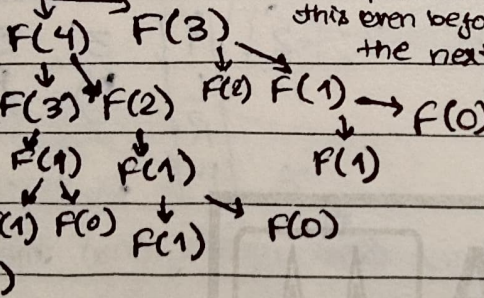
base cases are required

{ if $n==0$: return 0.

elif $n==1$: return 1

else: return $F(n-1)+F(n-2)$ } Recursive part

$F(5)$



$|F(5)=5| \leftarrow$

* Recursion uses a

Recursive call Stack

Time: $O(2^n)$

Space: $O(n)$

used commonly in

- Linked Lists
- Trees
- Graphs

